

CoAP: Get Started with IoT Protocols

CoAP is a software protocol that allows simple electronic devices to communicate over the Internet. It is designed for small devices with low-power sensors and actuators that need to be controlled or supervised remotely, through standard Internet networks.



Network protocols play a significant role in communicating between various building blocks of an IoT architecture. You might wish for an efficient protocol that connects gateways with sensor nodes as per M2M or WSN needs, and with servers or cloud platforms for Web integration and global access. HTTP may be fit for certain needs but is expensive and has its own overheads. CoAP is a simple, less expensive protocol that meets all the above needs and is affordable for those with resource constraints.

Constrained Application Protocol (CoAP) is a simple, low overhead protocol designed for environments like low-end microcontrollers and constrained networks with critical bandwidth and a high error rate such as 6LoWPANs. It is defined by the IETF open standard RFC 7252, runs on UDP by default but is not limited to it, as it can be implemented over other channels like TCP, DTLS or SMS. A typical networking stack offering CoAP is shown in Figure 1. CoAP is based on the request-response communication model and comes with support for resource discovery, better reliability, URIs, etc. The protocol was designed for M2M needs initially, but was adapted in IoT as well, with support on gateways, high-end servers and enterprise integration.

CoAP resembles HTTP in terms of the REST model with *GET*, *POST*, *PUT* and *DELETE* methods, URIs, response codes, MIME types, etc, but one shouldn't think of it as compressed HTTP. However, CoAP can easily interface with HTTP using proxy components, where HTTP clients can talk

to CoAP servers and vice versa, which enables better Web integration and the ability to meet IoT needs.

Let's have a quick overview of the protocol.

Endpoints

A host or node participating in CoAP communication is known as an endpoint. The endpoint on which resources are defined and is a destination for requests is known as a server or, more precisely, the origin server and the endpoint from which requests are made for target resources is known as a client. Similarly, a server is the source and a client is the destination for responses. Certain endpoints like proxies act as the intermediate client and server.

The CoAP message format

One of the key design goals of CoAP is to avoid fragmentation at underlying layers, especially at the link layer, i.e., the whole CoAP packet should fit into a single datagram compatible with a single frame at the Ethernet or IEEE 802.15.4 layer. This is possible with a compact 4-byte binary header, optional fields and payload, as shown in Figure 2.

Version: 2-bit version number, currently fixed at 0x01

Type of messages: CoAP supports four types of messages with the following 2 bit transaction codes:

CON	00(0)	ACK	10(2)
NON	01(1)	RST	11(3)

CoAP offers optional reliability using conformable